

Leonardo Herrera

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Spring, TX

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Portfolio: herrera2.github.io/Personal-Website/index.html

LinkedIn: linkedin.com/in/lh-mech-eng

Skills

SolidWorks, Creo Parametric, Siemens NX, AutoCAD, GD&T, Tolerance Stack-Up, Sheet Metal Design, CNC Nesting, MATLAB/Simulink, System Identification, Transfer Function Analysis, FEA, Hardware-in-the-Loop Simulation, Systems Integration, Sensor Calibration, Root Cause Analysis, Process Improvement, Trilingual (English, Spanish, Portuguese)

Work Experience

Manufacturing Design Engineer

May 2025 – Present

Duke Manufacturing – CounterCraft

Houston, TX

- Design and release ~70 custom sheet-metal components daily in SolidWorks for made-to-order commercial countertop systems, contributing to the delivery of 100+ custom counter assemblies over a 10-month period.
- Generate flat patterns, bend deductions, and CNC-ready nesting programs for ~200 parts per day, improving material utilization, reducing scrap, and maintaining high-throughput manufacturing schedules.
- Developed and implemented a parametric universal mullion framework in SolidWorks that standardized internal support structures across product lines, reducing modeling and design iteration time by ~15–20%.
- Collaborate with fabrication, CNC, and production teams to resolve fit-up and structural nonconformances, reducing production issues and improving manufacturing flow.
- Interpret and review engineering drawings in AutoCAD prior to modeling, identifying critical design discrepancies that prevented significant manufacturing rework, stainless-steel scrap, and production delays.

Mechanical & Test Engineering Intern

March 2022 – September 2023

NASA Johnson Space Center – CACI (Contractor), PVAMU (Subcontractor)

Houston, TX

- Supported spacecraft docking certification campaigns comprising 300+ dynamic test cases on the [SDTS](#), contributing to hardware setup, calibration, and execution of real-time hardware-in-the-loop simulations.
- Integrated and configured electromechanical instrumentation, optical tracking systems, and laser metrology equipment to support accurate multi-system validation during spacecraft docking tests.
- Operated a 6DOF robotic simulation system in closed-loop docking tests, using real-time sensor feedback to dynamically adjust vehicle motion and analyze system performance during contact events.
- Designed and improved test hardware, including sensor mounts and structural interfaces, using Creo Parametric and Creo Simulate to support hardware integration and docking test operations.
- Developed calibration tooling and validated force-measurement systems to improve sensor accuracy, data consistency, and test repeatability across docking campaigns.
- Built dynamic system models in MATLAB/Simulink, applying system identification and transfer-function analysis to characterize robotic system behavior and support control-filter assessment studies.
- Collaborated with multidisciplinary engineering teams supporting mission-critical validation efforts for international and commercial partner vehicles, including SpaceX and Japan Aerospace Exploration Agency.
- Recognized with two NASA Elite Team Awards and two CACI Recognitions for contributions to spacecraft docking simulation and SDTS test operations.

Engineering Projects

Robotic Arm Simulation (NASA Trick) (*In Development*)

May 2026 – Present

- Developing a 3-DOF robotic arm simulation in NASA Trick using Python/C++, implementing forward and inverse kinematics for real-time motion control, trajectory visualization, and dynamic simulation.

Stewart Platform Kinematics & Simulation (stewart_platform.html)

March 2026 – April 2026

- Developed inverse and forward kinematics for a 6-DOF Stewart platform, solving constrained actuator motion through numerical optimization and enabling interactive real-time 3D visualization.

Education

Master of Science in Engineering

Graduated December 2023

Prairie View A&M University, Prairie View, TX

GPA: 3.80

- Thesis: Control Filter Assessment for a Six Degree-of-Freedom Stewart Platform (Publication Pending)

Bachelor of Science in Mechanical Engineering

Graduated May 2022

Prairie View A&M University, Prairie View, TX

GPA: 3.67

- Senior Project: Designed, modeled, and prototyped a [Lightweight Deployable Shield](#) for Ball Aerospace.